Incomplete information games

Incomplete information

- In a game of incomplete information, players may be uninformed about certain characteristics of the game or of the players.
- For instance, a player may have incomplete information about actions available to some other player, or about payoffs of other players, or about what other players know about him.
- In this course, we will study the first incomplete information model, due to John Harsanyi (1967).

Idea of the model

- Incomplete information is modelled by assuming that every player can be of a number of different types.
- A type of a player summarizes all relevant information (in particular, actions and payoffs) about that player.
- Furthermore, it is assumed that each player knows his own type and, given his own type, has a probability distribution over the types of the other players.
- It is also assumed that these probability distributions are consistent in the sense that they are the marginal probability distributions derived from a basic, commonly known distribution over all combinations of player types.

Idea of the model

- As before, we will only consider finite games, meaning there are finitely many players with finitely many types, and at each type, there are finitely many actions.

Example

- Consider a variant of the Battle-of-the-Sexes in which Player 1 (the man) does not know whether Player 2 (the woman) wants to go out with him or avoid him.
- More precisely, the man does not know whether he plays the game y (from "yes") or the game n (from "no")

$$y: \begin{array}{c c} & \begin{array}{cc} F & B \end{array} \\ \begin{array}{c} F \\ B \end{array} & \begin{pmatrix} 2,1 & 0,0 \\ 0,0 & 1,2 \end{pmatrix} \end{array} \quad n: \begin{array}{c c} & \begin{array}{cc} F & B \end{array} \\ \begin{array}{c} F \\ B \end{array} & \begin{pmatrix} 2,0 & 0,2 \\ 0,1 & 1,0 \end{pmatrix} \end{array}$$

Figure: Figure 1

Example

- Player 1 attaches probability $\frac{1}{2}$ to each of these games, and this is common knowledge.
- In the terminology of types, this means that Player 1 has only one type, simply indicated by "1", and that player 2 has two types, namely y and n.
- So there are two type combinations, namely $(1, y)$ and $(1, n)$, each occurring with probability $\frac{1}{2}$.
- Player 2 knows Player 1's type with certainty, and also knows her own type, that is, she knows which game is actually being played.
- Player 1 attaches probability $\frac{1}{2}$ to each type of Player 2.

Example

- Observe that here Player 1 has incomplete information about Player 2's utility.
- Similarly, we can have examples where a player has incomplete information about other player's available strategies, and as a result has incomplete information about his own utilities.

Formal model

Definition

A Harasanyi game with incomplete information is a vector $(N, (T_i)_{i \in N}, p, S, (s_t)_{t \in \times_{i \in N} T_i})$ where

- N is a finite set of players.
- T_i is a finite set of types. The set of type vectors is denoted by $T = \times_{i \in N} T_i$.
- $p \in \Delta T$ is a probability distribution over the set of type vectors that satisfies $p(t_i) = \sum_{t_{-i} \in T_{-i}} p(t_i, t_{-i}) > 0$ for every player $i \in N$ and every type $t_i \in T_i$.

Formal model

Definition

- S is a set state games.15 Every state game $s \in S$ is a vector $s = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$, where A_i is a nonempty set of actions of Player i and $u_i : \times_{i \in N} A_i \to \mathbb{R}$ is the payoff function of Player i.
- $s_t = (N, (A_i(t_i))_{i \in N}, (u_i(t))_{i \in N})$ is the state game for the type vector t, for every $t \in T$. Thus, Player i's action set in the state game s_t depends on his type t_i only, and is independent of the types of the other players.

How the game proceeds

- A chance move selects a type vector $t = (t_1, \ldots, t_n) \in T$ according to the probability distribution p.
- Every Player i knows the type t_i that has been selected for him (i.e., his own type), but does not know the types $t_{-i} = (t_j)_{j \neq i}$ of the other players.
- The players select their actions simultaneously: each Player i, knowing his type t_i, selects an action $a_i \in A_i(t_i)$.
- Every Player i receives the payoff $u_i(t; a)$, where $a = (a_1, a_2, \ldots, a_n)$ is the vector of actions that have been selected by the players.

Discussion

- Player i, of type t_i, does not know the types of the other players; he has a belief about their types.
- This belief is the conditional probability distribution $p(\cdot|t_i)$ over the set $T_{-i} = \times_{j \neq i} T_j$ of the vectors of the types of the other players.
- The set of actions that Player i believes that he has at his disposal is part of his type, and therefore the set $A_i(t_i)$ of actions available to Player i in the state game st is determined by t_i only, and not by the types of other players.

Discussion

- However, the utility of Player i depends on the type vector as well as the actions chosen by all the players.

Example

- Let's again go back to the variant of the battle of the sexes game.
- There are two players, a man and a woman. The man has one type: 1, and the woman has two types: y and n.
- So, two possible type combinations, $(1, y)$ and $(1, n)$.
- p is a probability distribution over the types, with each type having equal probabilities.

Example

- Further, there are two state games as shown in Figure 1.
- The type profile $(1, y)$ is assigned to the game in the left, and the type profile $(1, y)$ is assigned to the game in the right.
- The conditional distribution of Player 1 (the man) given his own type is

$$p(y|1) = p(n|1) = \frac{1}{2}.$$

Example

- Observe that the game can be written as an extensive form game with a chance node and imperfect information.

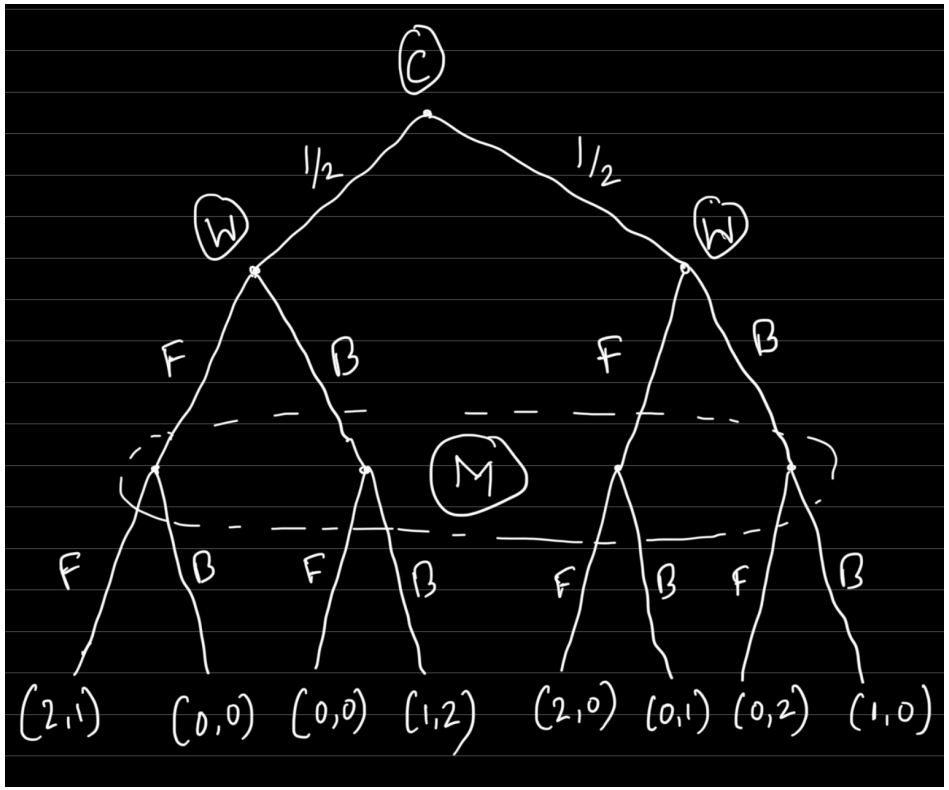

Figure: Figure 2

Example

- Similarly, we may consider a generalization of this variant where both men and women have two types: yes or no.
- There will be four type profiles now: (y_M, y_W), (y_M, n_W), (n_M, y_W), and (n_M, n_W).
- p is a probability distribution over these four type profiles.
- We assume that p is common knowledge. And, both men and women know their types and can compute the conditional probability over the other players' types.

Example

- Moreover, this game can also be written in an extensive form game in the following way.

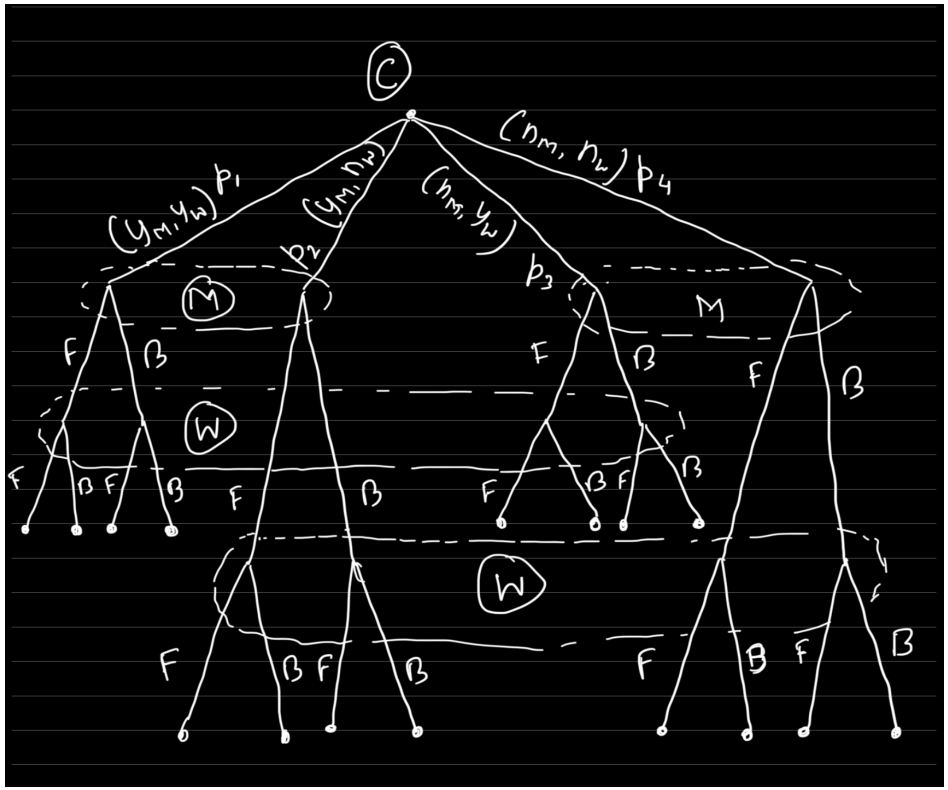

Figure: Figure 3

Theorem

- The previous observations holds in general as well. And, we have the following theorem.

Theorem

Every (Harsanyi) game with incomplete information can be described as an extensive-form game (with moves of chance and information sets).

Observation

- Note that in such a game, each player has as many information sets as his number of types.
- Further, the game has perfect recall, implying it is enough to consider behavior strategies, only.

Discussion on p

- Note that a crucial assumption in the model is that p is common knowledge.
- Another way of saying this is that every player's conditional distributions are calculated from a commonly known distribution over all the type profiles.
- One could also see this as something that is known from past incidents (see the next example).

Example

- Suppose in the town of Smallville, people regard the characteristic of being a family person or being a careerist person quite important for marriage.
- A local matchmaker Kevin approaches two of the residents, Abe and Sarah, informing them that in his opinion they would be well suited as a couple.
- Abe and Sarah know whether they are family person or careerist person but does not know about the other person.

Example

- However, it is known in the town that Kevin tends to match a man who stresses his career with a woman who emphasizes family life, and a man who puts family first with a woman who is career-oriented, but there were several instances in the past when Kevin did not stick to that rule.
- The distribution of past matches initiated by Kevin is presented in the following diagram

	Family Woman	**Career Woman**
Family Man	$\frac{1}{10}$	$\frac{3}{10}$
Career Man	$\frac{4}{10}$	$\frac{2}{10}$

Example

- The above situation can be modeled as an incomplete information game between Abe and Sarah where the common distribution over the type profiles could be the distribution over past matches initiated by Kevin.
- Further, given this Abe can calculate the probability of Sarah's type given his own type using conditional probability,
- For example,

$$p(\text{Sarah is a careerist} \mid \text{Abe is a careerist}) = \frac{\frac{2}{10}}{\frac{2}{10}+\frac{4}{10}} = \frac{1}{3}.$$

Strategies

- A pure strategy of Player i is a function $s_i : T_i \to \cup_{t_i \in T_i} A_i(t_i)$ such that $s_i(t_i) \in A_i(t_i)$ for all $t_i \in T_i$.
- A mixed strategy is a probability distribution over the set of pure strategies.
- A behavior strategy $\sigma_i : T_i \to \cup_{t_i \in T_i} \Delta A_i(t_i)$ such that $b_i(t_i) \in \Delta A_i(t_i)$ for all $t_i \in T_i$.
- We will consider behavior strategies only, and the set of behavior strategies for Player i is denoted by $\mathcal{B}_i$.

Utilities

- If the vector of the players' behavior strategies is $\sigma = (\sigma_1, \ldots, \sigma_n)$, and the vector of types that is selected by the chance move is $t = (t_1, \ldots, t_n)$, then each vector of actions $(a_1, \ldots, a_n)$ is selected with probability

$$\sigma_1(t_1; a_1) \times \sigma_2(a_2; t_2) \times \cdots \times \sigma(a_n, t_n).$$

- Player i's expected payoff, which we denote by $U_i(t; \sigma)$, is therefore

$$U_i(t; \sigma) = \sum_{a \in \times_{i \in N} A_i(t_i)} \sigma_1(t_1; a_1) \times \cdots \times \sigma(a_n, t_n).$$

Utilities

- Finally, the utility of Player i when the implemented strategy profile is σ is

$$U_i(\sigma) = \sum_{t \in T} p(t) U_i(t; \sigma).$$

- We call this to be the expected utility of Player i at the ex ante stage, i.e., before the lottery (randomization) takes place.
- After the chance move (called the interim stage), the vector of types has been selected, and the conditional expected payoff to Player i of type t_i is

$$U_i(\sigma | t_i) = \sum_{t_{-i} \in T_{-i}} p(t_{-i} | t_i) U_i((t_i, t_{-i}); \sigma).$$

Utilities

- The connection between the ex ante (unconditional expected) payoff $U_i(\sigma)$ and the interim (conditional) payoff $(U_i(\sigma|t_i))_{t_i \in T_i}$ is given by the equation

$$U_i(\sigma) = \sum_{t_i \in T_i} p(t_i) U_i(\sigma|t_i).$$

Equilibrium

- Harsanyi game of incomplete information may be analyzed at two separate points of time; at the ex ante stage and at the interim stage.
- Accordingly, there can be two types of equilibrium we may think of; Nash Equilibrium and Bayesian Equilibrium.
- The first equilibrium concept, which is Nash equilibrium in Harsanyi games, poses the requirement that no player can profit by a unilateral deviation before knowing his type.
- The second equilibrium concept, called Bayesian equilibrium, poses the requirement that no Player i can profit by deviating at the interim stage, after learning his type t_i.

Nash Equilibrium

Definition

A strategy vector σ^* is a Nash equilibrium if for each Player i and each strategy σ_i of Player i,

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma^*_{-i}).$$

Existence

- As every game with incomplete information can be described as an extensive-form game, every finite extensive-form game has a Nash equilibrium in mixed strategies, and every mixed strategy is equivalent to a behavior strategy, we arrive at the following conclusion:

Theorem

Every game with incomplete information in which the set of types is finite and the set of actions of each type is finite has a Nash equilibrium in behavior strategies.

Bayesian Equilibrium

Definition

A strategy vector $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a Bayesian equilibrium if for each player $i \in N$, each type $t_i \in T_i$, and each possible action $a_i \in A_i(t_i)$,

$$U_i(\sigma^* | t_i) \geq U_i((a_i, \sigma_{-i}) | t_i).$$

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Existence of Bayesian Equilibrium

- It can be shown that a Bayesian equilibrium in an incomplete information game is equivalent to a Nash equilibrium in the agent-form game.
- As the agent-form game is a finite game, there always exists an NE. Therefore, we have the following theorem.

Theorem

Every game with incomplete information in which the set of types is finite and the set of actions of each type is finite has a Bayesian equilibrium (in behavior strategies).

Relationship between the two equilibrium notions

- As $U_i(\sigma) = \sum_{t_i \in T_i} U_i(\sigma|t_i)$, it is quite straightforward to see that Bayesian equilibrium implies Nash equilibrium.
- The intuition is that if an agent cannot do better after knowing their type for all possible types, as the utility at the ex ante stage is a convex combination of the utilities at different types, the agent cannot be better off by deviating at the ex ante stage as well.

Relationship between the two equilibrium notions

- A bit surprisingly, the opposite direction holds as well, i.e., every Nash equilibrium is also a Bayesian equilibrium.
- Here the idea of the proof is if an equilibrium is not a Bayesian equilibrium, it is not a Nash equilibrium either.
- The detailed proof can be found in Solan's book.

Relationship between the two equilibrium notions

- Thus, we have the following theorem.

Theorem (Harsanyi (1967))

In a game with incomplete information in which the number of types of each player is finite, every Bayesian equilibrium is also a Nash equilibrium, and conversely every Nash equilibrium is also a Bayesian equilibrium.

Example

- We now see an example to compute all the NE in an incomplete information game by computing the corresponding Bayesian equilibria.
  - $N = \{1, 2\}$
  - $T_1 = \{I_1, I_2\}$ and $T_1 = \{II\}$
  - $p(I_1, II) = p(I_2, II) = \frac{1}{2}$
  - the two state games are as follows:

Player II

Player I		L	R
	T_1	$(1, 0)$	$(0, 2)$
	B_1	$(0, 3)$	$(1, 0)$

The state game for $t = (I_1, II_1)$

Player II

Player I		L	R
	T_2	$(0, 2)$	$(1, 1)$
	B_2	$(1, 0)$	$(0, 2)$

The state game for $t = (I_1, II_2)$

Example

- Turning to the calculation of Bayesian equilibria in this game, given such an equilibrium, denote by $[q(L), (1-q)(R)]$ the equilibrium strategy of Player 2, by $[x(T_1), (1-x)(B_1)]$ the equilibrium strategy of Player 1 of type I_1, and by $[y(T_2), (1-y)(B_2)]$ the equilibrium strategy of Player 1 of type I_2.
- We first show that $0 < q < 1$.
- If $q = 1$, then type I_1's best reply is $T(x = 1)$ and I_2's best reply is $B(y = 0)$. But Player 2's best reply to this strategy is $R(q = 0)$. It follows that $q = 1$ is not part of a Bayesian equilibrium.

Example

- Similar arguments hold for $q = 0$.
- The conclusion is therefore that in a Bayesian equilibrium Player 2's strategy must be completely mixed, so that he is necessarily indifferent between L and R.
- This implies that

$$\frac{1}{2} \cdot 3(1-x) + \frac{1}{2} \cdot 2y = \frac{1}{2} \cdot 2x + \frac{1}{2}(y + 2(1-y)),$$

$$\implies x = \frac{1+3y}{5}.$$

Example

- On the other hand, for Player 1, (x, y) to be an equilibrium, it must be the best reply of q.
- Note that if $q < \frac{1}{2}$, Player 1's best reply is $x = 0$, $y = 1$, which does not satisfy $x = \frac{1+3y}{5}$.
- Similar thing holds for $q > \frac{1}{2}$.
- For $q = \frac{1}{2}$, Player 1's payoff is $\frac{1}{2}$ irrespective of what he plays, so that every pair (x, y) is a best reply to $q = \frac{1}{2}$. .

Example

- This leads to the conclusion that a pair of strategies $(x, y; q)$ is a Bayesian equilibrium if and only if $q = \frac{1}{2}$ and $x = \frac{1+3y}{5}$.
- Since x and y are both in the interval $[0, 1]$ we obtain

$$\frac{1}{5} \leq x \leq \frac{4}{5},\ 0 \leq y \leq 1,\ x = \frac{1+3y}{5}$$

- At such a equilibrium, Player 1's utility is $\frac{1}{2}$ for both his types, and for Player 2, the utility is

$$\frac{1}{2} \cdot 3(1 - x) + \frac{1}{2} \cdot 2y = \frac{12+y}{10}.$$